EX.NO: 06

Deluge Scripting in Zoho Creator

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AIM

To create and run ten Deluge scripts in Zoho Creator covering arithmetic, decisions, loops, strings, lists, maps, date handling and the Fibonacci sequence, and verify the displayed outputs.

THEORY

Deluge (Data Enriched Language for the Universal Grid Environment) is Zoho Creator's online scripting language. It adds logic to applications, for example by validating data, calculating values, retrieving or updating records, and connecting to other services. Scripts can run through workflow events or custom functions.

Variables are assigned values without a separate data-type declaration. Deluge supports arithmetic, relational and logical operators. Conditional logic uses if, else if and else, while for each iterates through items in a list or collection.

Lists keep ordered values and provide operations such as add, get, size, sort, removeElement and contains. Maps store key-value pairs using put and get. Built-in string and date functions include toUpperCase, length, subString, indexOf, replaceAll, toDate, addDay and toString. The info statement prints values to the output console and helps test scripts.

PROCEDURE

1. Sign in to Zoho Creator and open the application used for testing.

2. Open Workflow, select Functions, and create a custom function in the Deluge editor.

3. Enter each program in the script editor and use info statements to display results.

4. Save and run the function, then check the output console.

5. Record the output and capture a screenshot for each program.

6. Repeat the process for all ten programs.

PROGRAMS AND OUTPUT

Program 1: Arithmetic Operations

Aim: Perform basic arithmetic calculations using two values.

Program:

```
a = 25;
b = 7;
info "Sum = " + (a + b);
info "Difference = " + (a - b);
info "Product = " + (a * b);
info "Remainder = " + (a % b);
```

Output:

```
Sum = 32
Difference = 18
Product = 175
Remainder = 4
```

Program 2: Even or Odd Number

Aim: Determine whether an integer is even or odd using if-else.

Program:

```
num = 42;
if(num % 2 == 0)
{
info num.toString() + " is an Even number";
}
else
{
info num.toString() + " is an Odd number";
}
```

Output:

```
42 is an Even number
```

Program 3: Largest of Three Numbers

Aim: Find the greatest value among three numbers using conditional branches.

Program:

```
a = 34;
b = 78;
c = 56;
if(a > b && a > c)
{
largest = a;
}
else if(b > c)
{
largest = b;
}
else
{
largest = c;
}
info "Largest number = " + largest;
```

Output:

```
Largest number = 78
```

Program 4: Factorial of a Number

Aim: Calculate 5 factorial with a for-each loop.

Program:

```
fact = 1;
for each n in {1, 2, 3, 4, 5}
{
fact = fact * n;
}
info "Factorial of 5 = " + fact;
```

Output:

```
Factorial of 5 = 120
```

Program 5: Sum, Count and Maximum of a List

Aim: Calculate the total, number of elements and maximum value in a list.

Program:

```
numbers = {12, 45, 7, 89, 23};
total = 0;
maxVal = numbers.get(0);
for each num in numbers
{
total = total + num;
if(num > maxVal)
{
maxVal = num;
}
}
info "Sum = " + total;
info "Count = " + numbers.size();
info "Maximum = " + maxVal;
```

Output:

```
Sum = 176
Count = 5
Maximum = 89
```

Program 6: String Functions

Aim: Demonstrate useful built-in functions for text values.

Program:

```
text = "Zoho Creator";
info "Upper case : " + text.toUpperCase();
info "Lower case : " + text.toLowerCase();
info "Length : " + text.length();
info "Substring : " + text.subString(0, 4);
info "Index of Creator : " + text.indexOf("Creator");
info "Replaced : " + text.replaceAll("Creator", "Deluge");
```

Output:

```
Upper case : ZOHO CREATOR
Lower case : zoho creator
```

```
Length : 12
Substring : Zoho
Index of Creator : 5
Replaced : Zoho Deluge
```

Program 7: List Operations

Aim: Add, sort, remove an item from a list and check whether an item exists.

Program:

```
fruits = List();
fruits.add("Apple");
fruits.add("Mango");
fruits.add("Banana");
info "Total fruits : " + fruits.size();
fruits.sort(true);
for each fruit in fruits
{
info fruit;
}
fruits.removeElement("Mango");
info "After removing Mango : " + fruits.size();
info "Contains Apple : " + fruits.contains("Apple");
```

Output:

```
Total fruits : 3
Apple
Banana
Mango
After removing Mango : 2
Contains Apple : true
```

Program 8: Map (Key-Value) Operations

Aim: Save and retrieve student details using a map.

Program:

```
student = Map();
student.put("Name", "Umar");
student.put("Department", "Computer Science");
student.put("Marks", 85);
info "Name : " + student.get("Name");
info "Department : " + student.get("Department");
info "Marks : " + student.get("Marks");
if(student.get("Marks") >= 50)
{
info "Result : Pass";
}
else
{
info "Result : Fail";
}
```

Output:

```
Name : Umar
Department : Computer Science
Marks : 85
Result : Pass
```

Program 9: Date Functions

Aim: Use Deluge date functions to add days and format a date.

Program:

```
startDate = "20-Oct-2026".toDate();
endDate = startDate.addDay(10);
info "Start Date : " + toString(startDate, "dd-MMM-yyyy");
info "After 10 days : " + toString(endDate, "dd-MMM-yyyy");
info "Month : " + toString(endDate, "MMMM");
info "Day : " + toString(endDate, "EEEE");
```

Output:

```
Start Date : 20-Oct-2026
After 10 days : 30-Oct-2026
Month : October
Day : Friday
```

Program 10: Fibonacci Series

Aim: Generate the first ten Fibonacci terms with a list and loop.

Program:

```
fib = {0, 1};
for each i in {1, 2, 3, 4, 5, 6, 7, 8}
{
n = fib.size();
fib.add(fib.get(n - 1) + fib.get(n - 2));
}
series = "";
for each x in fib
{
series = series + x.toString() + " ";
}
info "Fibonacci Series : " + series;
```

Output:

```
Fibonacci Series : 0 1 1 2 3 5 8 13 21 34
```

RESULT

Thus, ten Deluge programs covering calculations, conditional statements, iteration, text functions, lists, maps, date operations and the Fibonacci sequence were prepared and executed in Zoho Creator, and the expected outputs were observed.

EX.NO:06 Deluge Scripting in Zoho Creator

ROLL NO :240701573

NAME: UMAR FAROOK K

AIM:

To write and execute ten programs using Deluge, the scripting language of Zoho Creator, covering variables, operators, conditional statements, loops, strings, lists, maps and dates, and to observe the output of each program.

THEORY:

Deluge. Deluge (Data Enriched Language for the Universal Grid Environment) is the online scripting language of Zoho Creator. It is used to add logic to applications, such as validating data, calculating field values, fetching and updating records, and integrating with other services. Scripts are written in the Deluge editor and are executed when a workflow event occurs or when a custom function is run.

Variables and operators. Variables are created by assigning a value (for example `a = 25;`) and need no data type declaration. Deluge supports arithmetic operators (+, -, *, /, %), relational operators (==, !=, <, >, <=, >=) and logical operators (&&, ||, !).

Control statements. Decisions are made using `if`, `else if` and `else`. Repetition is done with `for each`, which visits every element of a list or collection.

Collections. A *List* stores an ordered set of values and supports `add()`, `get()`, `size()`, `sort()`, `removeElement()` and `contains()`. A *Map* stores key-value pairs and uses `put()` and `get()`.

Built-in functions. String functions such as `toUpperCase()`, `length()`, `subString()`, `indexOf()` and `replaceAll()`, and date functions such as `toDate()`, `addDay()` and `toString(date, format)` are available.

The info statement. The `info` statement prints a value to the output console. It is used to display results and to debug scripts while testing.

PROCEDURE:

1 Sign in to Zoho Creator and open the application in which the scripts are to be tested.

2 Go to the **Workflow** tab, open **Functions** and create a new custom function using the Deluge editor.

3 Type the Deluge program in the script editor, using `info` statements to display the results.

4 Save the function and execute the script. The output appears in the output console of the editor.

5 Note the output and take a screenshot of it.

6 Repeat steps 3 to 5 for each of the ten programs given below.

240701573

PROGRAMS AND OUTPUT:

Program 1: Arithmetic Operations

Aim: To perform basic arithmetic operations on two numbers.

Program:

```
a = 25;
b = 7;
info "Sum = " + (a + b);
info "Difference = " + (a - b);
info "Product = " + (a * b);
info "Remainder = " + (a % b);
```

Output:

Output

```
Sum = 32
Difference = 18
Product = 175
Remainder = 4
```

Program 2: Even or Odd Number

Aim: To check whether a given number is even or odd using if-else.

Program:

```
num = 42;
if(num % 2 == 0)
{
    info num.toString() + " is an Even number";
}
else
{
    info num.toString() + " is an Odd number";
}
```

Output:

Output

```
42 is an Even number
```

Program 3: Largest of Three Numbers

Aim: To find the largest of three numbers using if - else if - else.

Program:

```
a = 34;
b = 78;
c = 56;
if(a > b && a > c)
{
    largest = a;
}
else if(b > c)
{
    largest = b;
}
else
{
    largest = c;
}
info "Largest number = " + largest;
```

Output:

Output

```
Largest number = 78
```

Program 4: Factorial of a Number

Aim: To find the factorial of 5 using a for each loop.

Program:

```
fact = 1;
for each n in {1, 2, 3, 4, 5}
{
    fact = fact * n;
}
info "Factorial of 5 = " + fact;
```

Output:

Output

```
Factorial of 5 = 120
```

Program 5: Sum, Count and Maximum of a List

Aim: To find the sum, count and maximum value of the numbers in a list.

Program:

```
numbers = {12, 45, 7, 89, 23};
total = 0;
maxVal = numbers.get(0);
for each num in numbers
{
    total = total + num;
    if(num > maxVal)
    {
        maxVal = num;
    }
}
info "Sum = " + total;
info "Count = " + numbers.size();
info "Maximum = " + maxVal;
```

Output:

Output

```
Sum = 176
Count = 5
Maximum = 89
```

Program 6: String Functions

Aim: To use built-in string functions of Deluge.

Program:

```
text = "Zoho Creator";
info "Upper case : " + text.toUpperCase();
info "Lower case : " + text.toLowerCase();
info "Length : " + text.length();
info "Substring : " + text.subString(0, 4);
info "Index of Creator : " + text.indexOf("Creator");
info "Replaced : " + text.replaceAll("Creator", "Deluge");
```

Output:

Output

```
Upper case : ZOHO CREATOR
Lower case : zoho creator
Length : 12
Substring : Zoho
Index of Creator : 5
Replaced : Zoho Deluge
```

Program 7: List Operations

Aim: To create a list and perform add, sort, remove and search operations.

Program:

```
fruits = List();
fruits.add("Apple");
fruits.add("Mango");
fruits.add("Banana");
info "Total fruits : " + fruits.size();
fruits.sort(true);
for each fruit in fruits
{
    info fruit;
}
fruits.removeElement("Mango");
info "After removing Mango : " + fruits.size();
info "Contains Apple : " + fruits.contains("Apple");
```

Output:

Output

```
Total fruits : 3
Apple
Banana
Mango
After removing Mango : 2
Contains Apple : true
```

Program 8: Map (Key-Value) Operations

Aim: To store and retrieve student details using a Map.

Program:

```
student = Map();
student.put("Name", "Umar");
student.put("Department", "Computer Science");
student.put("Marks", 85);
info "Name : " + student.get("Name");
info "Department : " + student.get("Department");
info "Marks : " + student.get("Marks");
if(student.get("Marks") >= 50)
{
    info "Result : Pass";
}
else
{
    info "Result : Fail";
}
```

Output:

Output

```
Name : Umar
Department : Computer Science
Marks : 85
Result : Pass
```

Program 9: Date Functions

Aim: To perform date calculations and formatting using Deluge date functions.

Program:

```
startDate = "20-Oct-2026".toDate();
endDate = startDate.addDay(10);
info "Start Date : " + toString(startDate, "dd-MMM-yyyy");
info "After 10 days : " + toString(endDate, "dd-MMM-yyyy");
info "Month : " + toString(endDate, "MMMM");
info "Day : " + toString(endDate, "EEEE");
```

Output:

Output

```
Start Date : 20-Oct-2026
After 10 days : 30-Oct-2026
Month : October
Day : Friday
```

Program 10: Fibonacci Series

Aim: To generate the first 10 terms of the Fibonacci series using a list and a loop.

Program:

```
fib = {0, 1};
for each i in {1, 2, 3, 4, 5, 6, 7, 8}
{
    n = fib.size();
    fib.add(fib.get(n - 1) + fib.get(n - 2));
}
series = "";
for each x in fib
{
    series = series + x.toString() + " ";
}
info "Fibonacci Series : " + series;
```

Output:

Output

```
Fibonacci Series : 0 1 1 2 3 5 8 13 21 34
```

RESULT:

Thus, ten Deluge programs covering arithmetic operations, conditional statements, loops, string functions, lists, maps, date functions and the Fibonacci series were written and executed in Zoho Creator, and the expected output was obtained for each program.